# Digital System Fundamentals

(CEIP Digital 2025)

Sorayut Glomglome

# Lecturer

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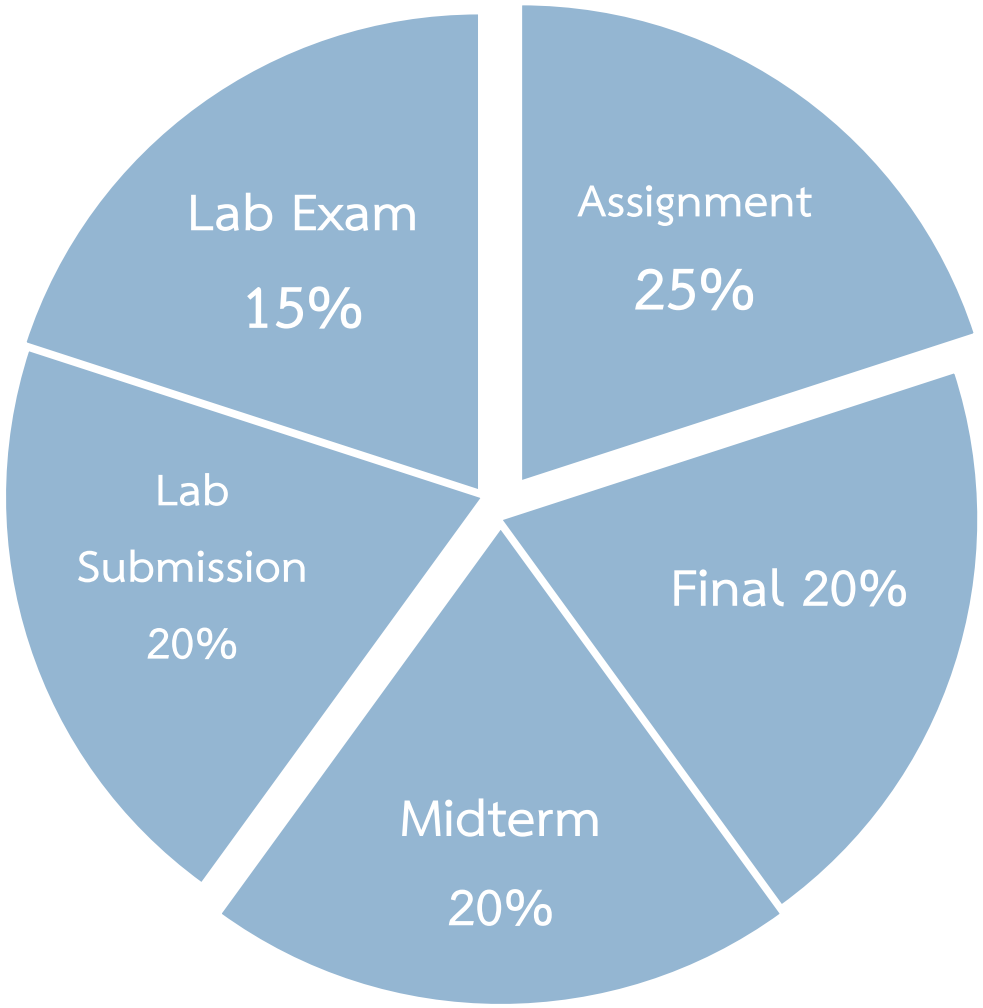
# Course Description

## 01276112 Digital System Fundamentals 4 (3-3-8)

- Introduction to digital systems
- Binary systems
- Boolean algebra and simplification
- Combinational circuit
- Sequential components, i.e., Latches, Flip-Flops, Registers, and Counters
- Sequential circuits
- Basic ALU (arithmetic and logic unit) and control unit
- Hardware description language

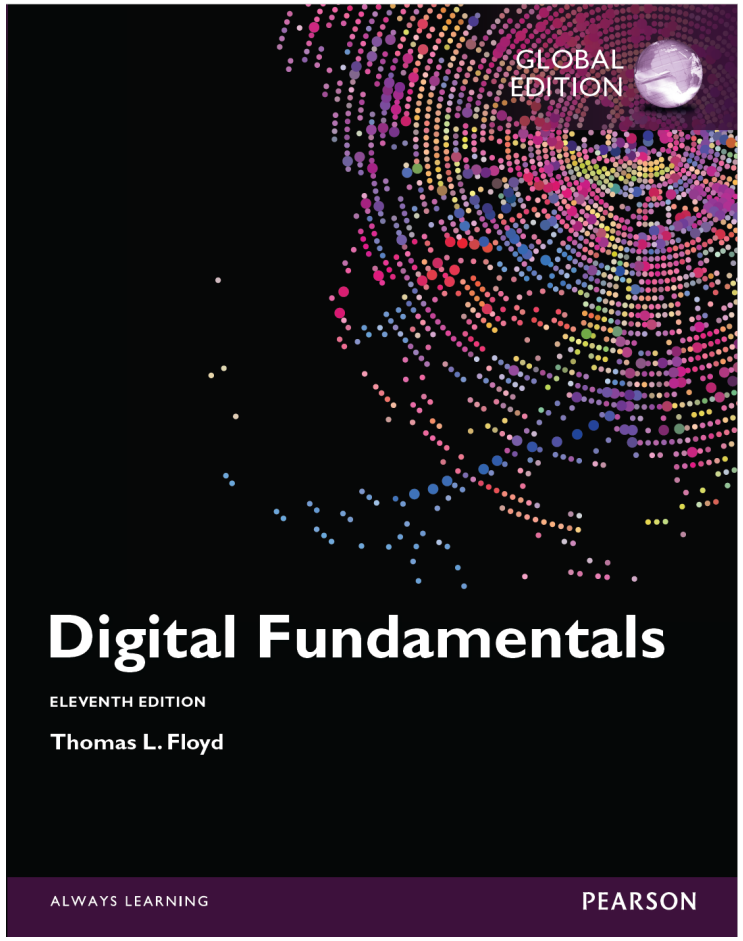
# Grade : Norm-Referenced Grading (อิงกลุ่ม)




| | |
|---|---|
| Lab | Group of Three |
| Lab Exam | Solo |
| Assignment | Two Lab Groups |
| Point >= 40 | Pass Grades |

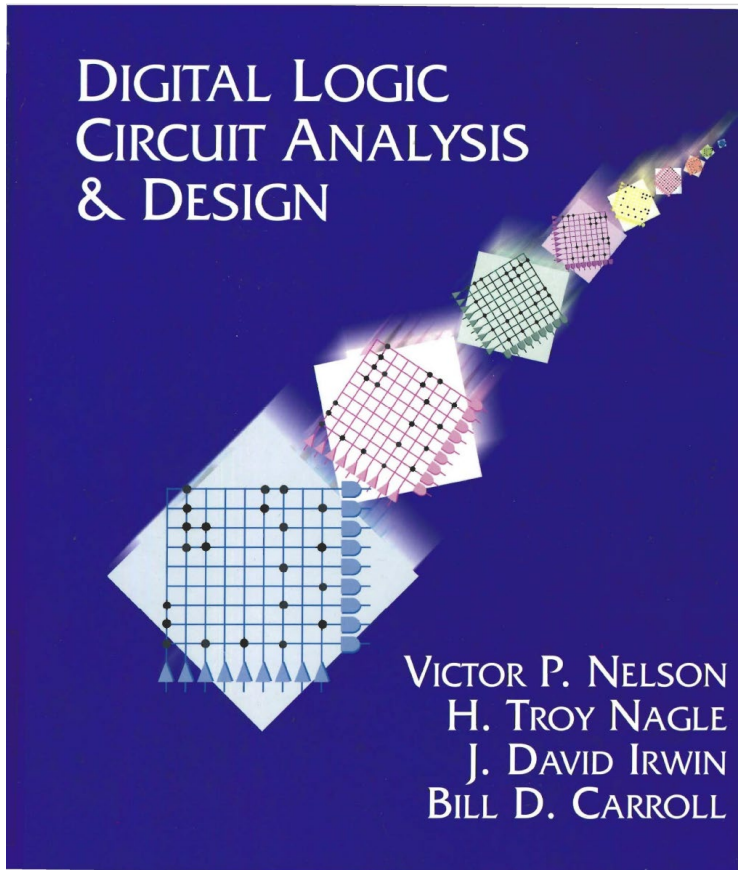
# Text

Thomas L. Floyd, **Digital Fundamentals,** 11th Edition**,** 2015



# Text

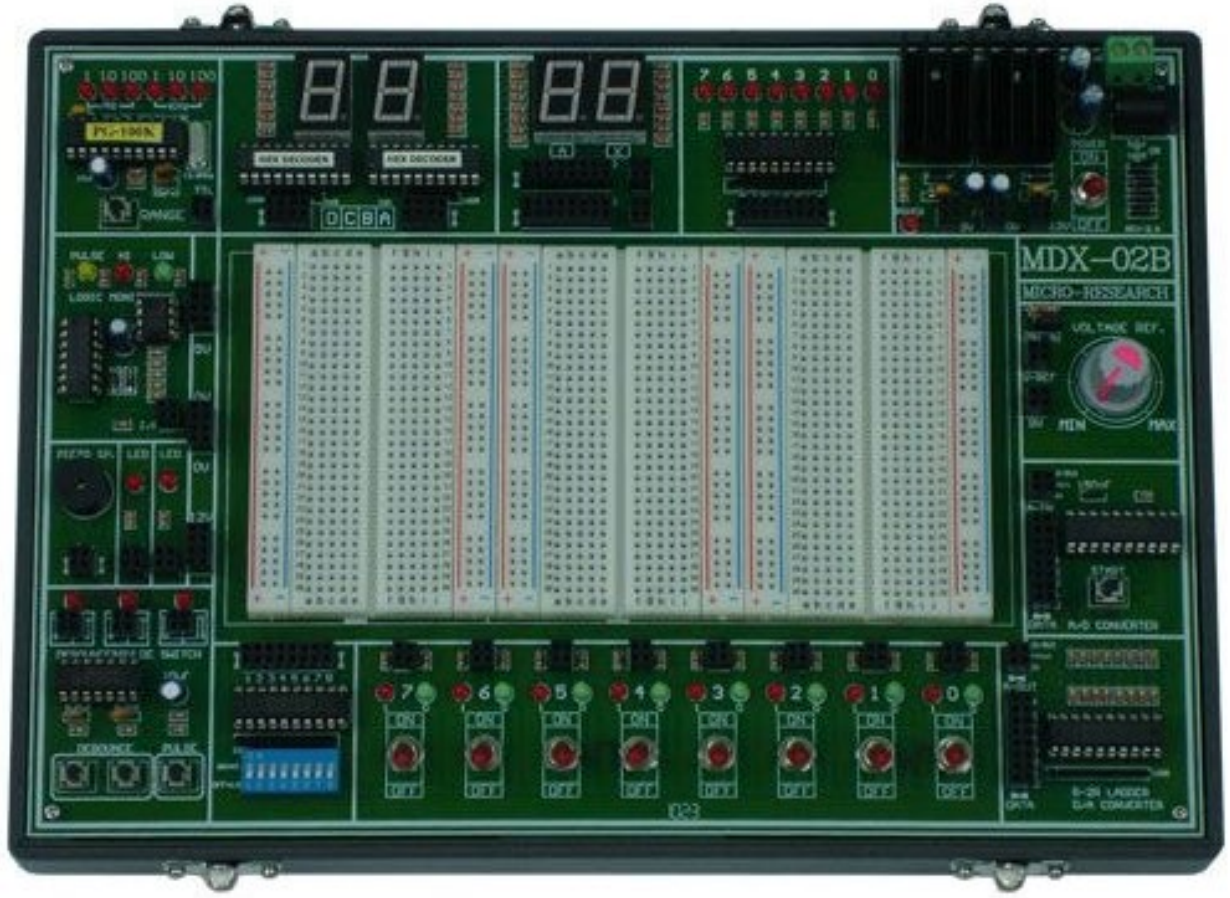
Victor P. Nelson, **Digital Logic Circuit Analysis & Design,** 1995



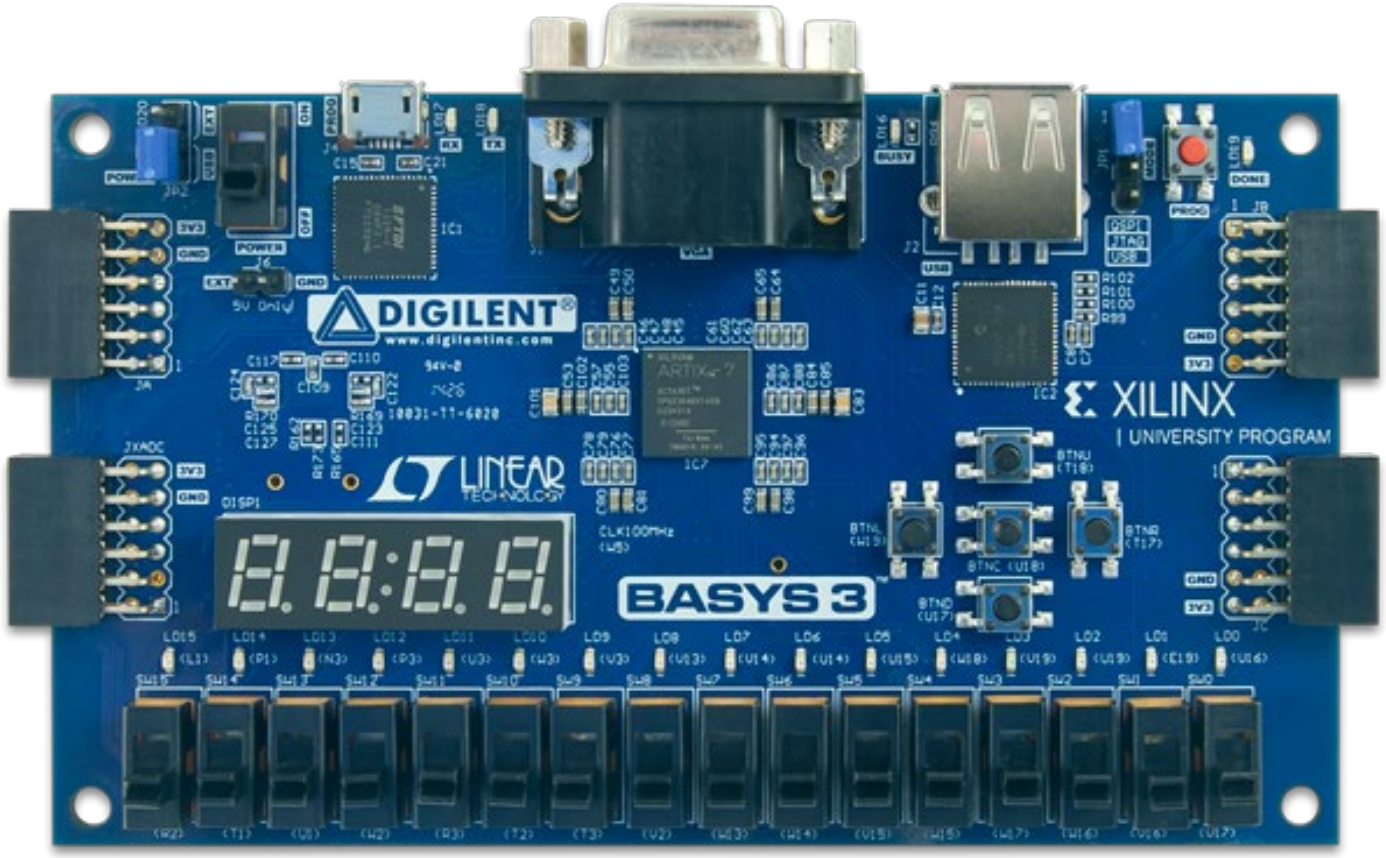
# Syllabus

| | Lecture | | Laboratory | |
|---|---|---|---|---|
| 27-Nov-25 | 1 & 2 | Introduction & Number Systems | | |
| 04-Dec-25 | 3 | Logic Gates | 1 | LogiSim |
| 11-Dec-25 | 4 | Boolean Algebra | 2 | Logic Gates |
| 18-Dec-25 | 5 | Quine-McClusky | 3 | Boolean Simplificati |
| 25-Dec-25 | 6 | Combination Circuit | 4 | Quine-McClusky |
| 01-Jan-26 | | No Class | | |
| 08-Jan-26 | | No Class | | |
| 15-Jan-26 | 7 | Latches & Flip-flips | 5 | Combination Circuit |
| 22-Jan-26 | Midterm | | | |
| 29-Jan-26 | 8 | Registers | 6 | FPGA-Adder |
| 05-Feb-26 | 9 | Counters | 7 | FPGA-Combination |
| 12-Feb-26 | 10 | Programable Logic D | 8 | FPGA-Register |
| 19-Feb-26 | 11 | Memory | 9 | FPGA-Counter |
| 26-Feb-26 | 12 | Arithmetic Logic Unit | | |
| 05-Mar-26 | | | | |
| 12-Mar-26 | | | | Lab Exam |
| 19-Mar-26 | Final | | | |
| 26-Mar-26 | Final | | | |
| 02-Apr-26 | Assignment Demonstration | | | |

# Digital Logic Trainer



# Basys 3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